DONALD AADITHIYAN

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Wellewatte, Colombo - 06, Sri Lanka

Summary

Final-year Computer Science undergraduate (Data Science specialization) at the University of Moratuwa. My research centers on mechanistic interpretability, using linear probing, activation editing, and causal tracing to recover latent features that neural networks encode internally but never express behaviorally. I'm particularly interested in how models reason and why they fail, and in using that understanding to build foundation models that not only perform better, but are safer and better aligned with human intent, rather than just observing them. I am also beginning to extend this work to vision-language models.

Education

- **University of Moratuwa** *Mar 2022 – Present*
  Bachelor of Science (Honours) in Engineering Moratuwa, Sri Lanka
  - Specialization: Computer Science and Engineering
  - Stream: Data Science and Engineering
  - Current CGPA: 3.56/4.0
- **St. Peter's College** *Jan 2019 – Jan 2022*
  GCE Advanced Level Colombo, Sri Lanka
  - Subjects: Distinction(A) in Physics, Chemistry, Combined Mathematics
  - Z-score: 2.2108 (Top 1.67% in Sri Lanka)

Publications

C=Conference, W=Workshop (Peer-Reviewed), WP=Working Paper (Not Peer-Reviewed)

[WP.1] **What Drives the Behavior?: A General Interpretability Pipeline for Finding Causal Features Behind Multi-Token Language-Model Behaviors**

[W.1] Donald Aadithiyan (2026). **Reading the Room: Implicit Confusion Encoding in Recurrent World Model States**. *GlobalSouthAI D&I Workshop, IJCAI-ECAI 2026*, Bremen, Germany. (Accepted - Presentation) [Paper]

[W.2] Donald Aadithiyan (2026). **Reaching the Tail: Calibration Diversity Drives Conformal Coverage under Data Scarcity**. *GlobalSouthAI D&I Workshop, IJCAI-ECAI 2026*, Bremen, Germany. (Accepted - Presentation) [Paper]

[C.1] Donald Aadithiyan, Birunthaban Rajendram, and Parishith Ragumar. (2025). **Predicting YouTube Video Initial Engagement Using Ensemble Machine Learning Techniques**. In *Proceedings of the International Conference in Data Science 2025*. Department of Statistics, University of Colombo, Sri Lanka. [Paper]

Research Experience

- **Xu Lab - Carnegie Mellon University & Harvard Medical School** *Jul 2026 – Present*
  Research Collaborator Lab Website
  - Building a deep learning pipeline with Dr. Xiangrui Zeng to improve temporal consistency of cortical thickness measurements across longitudinal MRI timepoints.
  - Handling ADNI preprocessing and MRI–PET integration, with a focus on layer-specific cortical segmentation (supragranular and infragranular).

Professional Experience

- **Differential Capital PTY LTD** *Dec 2025 – Jun 2026*
  Data Science Intern & Research Collaborator Differential Capital
  - Built a LightGBM-based machine valuation model on Refinitiv DataStream fundamentals for JSE equities, independently mapping 200+ features from a published quant research framework.
  - **Research:** Developed CAE-KAN for interpretable equity valuation on the JSE, introducing a Valuation Curvature Signal (VCS) for per-factor regime attribution. (Working paper)
- **Rising Talent Freelance Full Stack Developer – Upwork** *Jan 2025 – Feb 2026*
  Rising Talent Badge - Top 15% in the Industry Upwork Profile
  - Delivered two production full-stack applications (Flutter, Next.js, Firebase) for international clients with 5-star feedback.

Skills

- **Mechanistic interpretability & representation analysis :** linear probing, activation editing, causal tracing, PCA-based direction analysis, representational similarity analysis (RSA)
- **Model control & behavior steering :** activation steering, ablation studies, empirical null testing, cross-domain generalization analysis
- **Model-based RL & uncertainty quantification :** DreamerV3 world-model latent analysis, ensemble disagreement estimation, conformal inference (adaptive, Mondrian)
- **Interpretable architecture design :** KAN (B-spline) decoders, sparse/conditional autoencoders
- **Computing :** Python, PyTorch, R, SQL, TypeScript, JavaScript, Java, C++, Dart, Flutter, Next.js, Git